

COMP9120 26s1 Final Practice

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W1 Introduction

Q1. MCQ

Which statement best distinguishes a database schema from a database instance?

- (A) The schema is the current set of tuples; the instance is the physical index.
- (B) The schema is the metadata/design; the instance is the data at a particular time.
- (C) Both are updated whenever a tuple is inserted.
- (D) The schema is only used by application code, not the DBMS.

Q2. True/False

A DBMS is responsible for enforcing declared integrity constraints and supporting controlled concurrent access.

Q3. Short Answer

Give two reasons why storing application data in a DBMS is preferable to storing it only in ad-hoc files.

Q4. MCQ

Which pair is most closely associated with OLTP workloads?

- (A) Long scans and historical aggregates.
- (B) Short transactions and many concurrent updates.
- (C) Offline model training and denormalised snapshots.
- (D) Manual spreadsheet import only.

Q5. Short Answer

Explain logical data independence in one or two sentences.

Q6. Table/Diagram

Complete a three-row table matching each stage to its output: conceptual design, logical design, physical design.

Q7. MCQ Semester 2 2022 Final (modified)

Which option is NOT a typical DBMS function?

- (A) Enforcing integrity constraints.
- (B) Handling concurrent access.
- (C) Providing secure access.
- (D) Automatically deciding the business meaning of every attribute.

Q8. Short Answer

A student says: "normalisation is performed by the DBMS at runtime". Correct this statement.

Q9. Calculation

A table has 12,000 tuples and a query reads all tuples once. If a page stores 120 tuples, estimate the number of data pages read, ignoring headers and indexes.

Q10. Long Question

For a library database, list three entity sets and two relationships that would likely appear in a conceptual model. State one business rule for each relationship.



W2 Conceptual Database Design

Q1. MCQ

In an ER model, total participation of Entity A in relationship R means:

- (A) every A must participate in at least one R.
- (B) every R has exactly one A.
- (C) A is a weak entity.
- (D) R has no attributes.

Q2. Short Answer

Differentiate a weak entity from a strong entity. Include how its key is formed.

Q3. Table/Diagram

Map the pattern Customer 1:N Order and Order 1:N weak OrderLine(lineNo, qty) into relations.

Q4. MCQ

Which mapping is normally used for a many-to-many relationship Enrols(Student, Unit) with attribute mark?

- (A) Put mark in Student only.
- (B) Put mark in Unit only.
- (C) Create Enrols(studentId, unitId, mark) with a composite key.
- (D) No table is needed.

Q5. True/False

A multivalued attribute should usually be mapped into a separate relation containing the owner key and the attribute value.

Q6. Short Answer

What is aggregation in EER modelling, and when is it useful?

Q7. Long Question Semester 2 2021 Final (modified)

Modified ER question: A property has a propertyNo, type and address. A worker has licenceNo, name, birthDate and is either domestic with qualification or international with visaType, but not both. Describe an EER design.

Q8. MCQ

If every unit must have exactly one lecturer and a lecturer may teach zero or many units, the relationship from Lecturer to Unit is:

- (A) 1:1 with total Lecturer.
- (B) 1:N with total Unit participation.
- (C) M:N with total Lecturer.
- (D) N:1 with no constraints.

Q9. Short Answer

Why should relationship attributes usually not be placed on just one participating entity in an M:N relationship?

Q10. Long Question

Design a compact ER model for recipes shown on TV programs: each program includes one or more recipes; recipes can appear on multiple programs; each chef can demo zero or more programs and each program has exactly one chef. State cardinalities and a relational mapping.



W3 Relational Model and Logical Design

Q1. MCQ

Which statement about foreign keys is true?

- (A) A foreign key must always be unique.
- (B) A foreign key may contain NULL values unless constrained otherwise.
- (C) Every table must have exactly one foreign key.
- (D) A foreign key uniquely identifies each row in its own table.

Q2. True/False

A relation may have multiple candidate keys, but only one primary key is chosen in the schema.

Q3. Short Answer Semester 2 2021 Final (modified)

Using Ward(wno,name,mno), Bed(wno,bno,type), Manager(mno,mname), give one likely foreign key and one likely composite key.

Q4. MCQ

For values 8, 12, NULL, what does SQL AVG(x) return?

- (A) 20/3
- (B) 10
- (C) NULL
- (D) 20

Q5. Calculation

Evaluate the SQL three-valued expression: (TRUE AND UNKNOWN) OR FALSE.

Q6. Short Answer

What is the logical pointer between two relations in the relational model?

Q7. Table/Diagram

List the effect of these attempted operations: insert child with missing parent; delete parent with existing child; insert parent with no child.

Q8. MCQ

Which SQL constraint is the most direct way to express that a non-primary attribute combination must be unique?

- (A) CHECK
- (B) UNIQUE
- (C) FOREIGN KEY
- (D) DEFAULT

Q9. Short Answer

Why is entity integrity stronger for primary key attributes than for ordinary UNIQUE attributes?

Q10. Long Question

Map this EER fragment: Category(cid,manufacturer) has disjoint subtypes RadioSatellite(frequency) and GPSSatellite(imageResolution). Technician(tid,name) can be trained in many categories and categories can train many technicians. Give relations.



W4 Relational Algebra and SQL

Q1. MCQ

In relational algebra, selection σ corresponds most closely to which SQL clause?

- (A) SELECT list
- (B) FROM
- (C) WHERE
- (D) GROUP BY

Q2. Short Answer

Write an RA expression for supplier ids that have no shipment using Supplier(sid,name) and Shipment(sid,partId,qty).

Q3. Calculation

Relations R(A,B) and S(B,C) have tuples R=(1,2),(2,3), S=(2,9),(4,8). What is R natural join S?

Q4. MCQ

Which operator is required to keep unmatched rows from both inputs?

- (A) inner join
- (B) natural join only
- (C) full outer join
- (D) projection

Q5. Short Answer Semester 2 2024 Final (modified)

Explain why $\pi_{A,C}(\sigma_{D=2}(R \bowtie_{A=E} S))$ may not be equivalent to $(\pi_{A,C} R) \bowtie_{A=E} (\sigma_{D=2} S)$.

Q6. Long Question

Given Hotel(hid,name), Room(hid,rno,cost), Booking(cid,hid,rno,dateFrom,dateTo), Customer(cid,name,gender,age), write SQL for customer ids who booked a room with maximal cost.

Q7. Short Answer

Give a SQL pattern for finding cities never affected by any flood using Cities(cityId) and FloodAffectedCities(cityId,floodId).

Q8. MCQ

Which RA operator naturally captures “appears in both before-1995 and after-2020 result sets”?

- (A) union
- (B) intersection
- (C) difference
- (D) rename

Q9. Table/Diagram

Fill a two-column mapping: RA projection, selection, join, set difference to their common SQL idiom.

Q10. Long Question

Student(sid,name,country) and Enrolled(sid,unit,semester) are natural full outer joined. Describe what happens to students with no enrolment and enrolments whose sid has no matching student.



W5 Advanced SQL

Q1. MCQ

Which clause filters groups after aggregation?

- (A) WHERE
- (B) HAVING
- (C) FROM
- (D) ORDER BY

Q2. Short Answer

Write SQL logic for products having the largest number of distinct suppliers, preserving ties.

Q3. MCQ

Which form is usually safest for anti-join semantics in the presence of NULLs?

- (A) NOT EXISTS
- (B) NOT IN without NULL checks
- (C) NATURAL JOIN
- (D) ORDER BY

Q4. Short Answer

Explain the difference between WHERE and HAVING using one example.

Q5. Long Question

Author(authorId,name,country) and Book(bookId,title,authorId,storeCountry). Write SQL for author names whose books are sold in the same country as the author.

Q6. True/False

Every view involving a GROUP BY is straightforwardly updatable.

Q7. Table/Diagram

Compare IN, EXISTS and JOIN for subquery use in three short rows.

Q8. MCQ

To find customers who bought both product P1 and product P2, a robust grouped SQL condition is:

- (A) COUNT(DISTINCT productId)=2 after filtering to P1/P2.
- (B) ORDER BY purchaseDate.
- (C) WHERE productId='P1' AND productId='P2'.
- (D) LIMIT 2.

Q9. Short Answer

What is a correlated subquery?

Q10. Calculation

In a group with values 5, NULL, 15, what are COUNT(*), COUNT(value), SUM(value), AVG(value)?



W6 Database Integrity

Q1. MCQ

When must a referential integrity constraint from Child.fk to Parent.pk normally be checked?

- (A) Insert/update child fk and delete/update parent pk.
- (B) Only when parent is inserted.
- (C) Only at database creation.
- (D) Never if fk is nullable.

Q2. Short Answer Semester 2 2024 Final (modified)

Does a foreign key always have to be NOT NULL? Does it have to be UNIQUE?

Q3. MCQ

Which constraint best enforces that salary must be greater than zero for each row?

- (A) CHECK(salary>0)
- (B) FOREIGN KEY(salary)
- (C) ON DELETE CASCADE
- (D) NATURAL JOIN

Q4. Short Answer Semester 2 2024 Final (modified)

Translate this assertion idea into English: no flight exists without at least one booked reservation.

Q5. Table/Diagram

List three possible actions when deleting a parent row that is referenced by child rows.

Q6. MCQ

To delete child rows automatically when a referenced parent row is deleted, the cleanest DB-level enforcement is:

- (A) BEFORE ROW trigger on unrelated SELECT statements.
- (B) A declarative foreign key with ON DELETE CASCADE.
- (C) A CHECK constraint on the parent row only.
- (D) Manual deletion in a separate uncommitted session.

Q7. True/False

Assertions are standard SQL-style global constraints, but not all commercial DBMSs implement CREATE ASSERTION.

Q8. Short Answer

Give one reason to prefer a declarative FK action over a hand-written trigger.

Q9. Calculation

Table Project has primary keys 101,102. Employee has projId values 101, NULL, 999. Which employee row violates the FK Employee.projId references Project.projId, assuming projId is nullable?

Q10. Long Question

Projects(projId) and Employees(empId,projId) where projId is FK. If deleting a project should delete all employees assigned only to that project, describe SQL-level enforcement choices.



W7 Introduction to Database Application Development

Q1. MCQ

Which practice best prevents SQL injection?

- (A) String concatenation with user input.
- (B) Prepared statements with bound parameters.
- (C) Giving the app DBA privileges.
- (D) Hiding errors only.

Q2. Short Answer

What should happen to a transaction if the second SQL statement in a money transfer fails after the first update succeeded?

Q3. Table/Diagram

Draw or describe the layers in a simple database-backed web application.

Q4. MCQ

Why use connection pooling?

- (A) To reuse expensive database connections safely.
- (B) To remove the need for SQL.
- (C) To disable transactions.
- (D) To force every query to scan tables.

Q5. Short Answer

Explain the N+1 query problem.

Q6. True/False

An ORM completely removes the need to understand SQL execution and transaction boundaries.

Q7. Long Question

Design a transaction for enrolling a student in a unit with a capacity limit. Mention checks, updates and concurrency concern.

Q8. MCQ

Least privilege for an application account means:

- (A) grant only permissions required by the app.
- (B) grant owner privileges to simplify debugging.
- (C) avoid authentication.
- (D) store passwords in source code.

Q9. Short Answer

When might AFTER ROW triggers be appropriate in application-backed systems?

Q10. Calculation

A batch update affects 5,000 rows. A ROW trigger fires once per row; a STATEMENT trigger fires once per SQL statement. How many firings for each?



W8 Schema Refinement and Normalisation

Q1. MCQ

For $R(A,B,C,D)$ with $F=A \rightarrow B, B \rightarrow C$, what is A^+ ?

- (A) A
- (B) A,B
- (C) A,B,C
- (D) A,B,C,D

Q2. Short Answer

Why is every relation with only two attributes in BCNF?

Q3. Long Question Semester 2 2021 Final (modified)

Modified FD task: For $R(A,B,C,D)$, $F=AB \rightarrow C, A \rightarrow D$. Find a candidate key and identify a BCNF violation.

Q4. MCQ

BCNF requires that for every nontrivial FD $X \rightarrow Y$:

- (A) Y is a primary key.
- (B) X is a superkey.
- (C) X has one attribute.
- (D) the relation has no foreign keys.

Q5. Short Answer

Name the three common anomalies normalisation aims to reduce.

Q6. Calculation

Given $F=A \rightarrow BC, C \rightarrow D$ on $R(A,B,C,D)$, compute A^+ .

Q7. Table/Diagram

Contrast 3NF and BCNF in two rows.

Q8. Short Answer

What is a lossless decomposition?

Q9. MCQ

If $orderId \rightarrow customerName$ and $(orderId,lineNo)$ is the key of OrderLine, the FD $orderId \rightarrow customerName$ suggests:

- (A) no redundancy risk.
- (B) a partial dependency/anomaly risk.
- (C) a foreign key must be NULL.
- (D) the table is automatically BCNF.

Q10. Long Question

Decompose $R(A,B,C,D)$ with violating FD $B \rightarrow C$ into BCNF, assuming R has candidate key AD and other FDs allow it.



W9 Transaction Management

Q1. MCQ

A schedule is conflict serializable iff its precedence graph:

- (A) has a cycle involving all transactions.
- (B) has any cycle.
- (C) is acyclic.
- (D) has no edges.

Q2. Short Answer

For $R1(X)$, $W2(X)$, $W1(Y)$, $R2(Y)$, draw the precedence edges and state serializability.

Q3. Long Question Semester 2 2024 Final (modified)

Modified schedule: $R1(x)$, $W1(x)$, $R2(x)$, $R3(z)$, $W1(y)$, $R2(y)$, $W3(y)$, $R4(x)$, $R1(z)$. Determine conflict-serializability.

Q4. MCQ

Which lock compatibility is normally allowed?

- (A) S with S
- (B) S with X
- (C) X with X
- (D) X with S

Q5. Short Answer

Explain strict two-phase locking.

Q6. True/False

Deadlock can be detected by looking for a cycle in the wait-for graph.

Q7. Table/Diagram

List four anomalies prevented or controlled by transaction isolation.

Q8. Calculation

Schedule $W1(A)$, $R2(A)$, $W2(B)$, $R1(B)$. What edges exist?

Q9. MCQ

Durability means:

- (A) committed effects survive crashes.
- (B) every query is fast.
- (C) all schedules are serial.
- (D) no locks are needed.

Q10. Short Answer

Why should application transactions be kept short?



W10 Quiz and Guest Lecture Integration

Q1. MCQ

Which answer strategy is best for a quiz statement containing “every”, “at most one”, and “must”?

- (A) Ignore quantifiers and use intuition.
- (B) Translate each quantifier into min/max cardinality or constraint.
- (C) Assume all relationships are M:N.
- (D) Draw only tables.

Q2. Short Answer

A quiz asks whether relation $R(A,B)$ can violate BCNF. Give the short proof pattern.

Q3. Table/Diagram

Create a mini checklist for converting an ER diagram to relations in a timed quiz.

Q4. MCQ

Which issue is most likely when translating a global SQL assertion to English?

- (A) Misreading NOT EXISTS as existence of bad rows rather than absence of violations.
- (B) Forgetting page size.
- (C) Drawing too many B+ tree leaves.
- (D) Choosing a font.

Q5. Short Answer

State a compact method for answering conflict serializability questions in a quiz.

Q6. Calculation

In a quiz on NULLs, evaluate $(\text{NULL} = \text{NULL}) \text{ OR } \text{TRUE}$ under SQL three-valued logic.

Q7. Long Question

A quiz asks whether deleting a strong entity affects weak entities. Give a general answer and caveat.

Q8. MCQ

Which pair is most often tested together in final-style design questions?

- (A) ER mapping and SQL DDL constraints.
- (B) Font choice and margins.
- (C) HTML routing and CSS.
- (D) GPU kernels and shaders.

Q9. Short Answer

Explain why quiz SQL queries asking for “highest” should preserve ties unless stated otherwise.

Q10. Long Question

Given a mixed quiz question containing ER cardinality, RA, FK and transaction parts, outline the order in which you would solve it.



W11 Storage and Indexing

Q1. Calculation

A page is 4096 bytes with 196 bytes header. Record size is 32 bytes and target occupancy is 90 percent. How many records per page?

Q2. Calculation Semester 2 2024 Final (modified)

Using the previous records/page value, estimate pages for 400,000 records.

Q3. MCQ

Which index structure is generally best for range queries?

- (A) B+ tree
- (B) hash index
- (C) unordered heap file only
- (D) no index ever

Q4. Short Answer

Differentiate clustered and unclustered indexes.

Q5. Table/Diagram

Compare heap, sorted and hashed file organisation for equality and range search.

Q6. MCQ

A composite B+ tree index on (dept, salary) can directly support which prefix predicate best?

- (A) dept='IT'
- (B) salary>100000 only
- (C) name='Ann'
- (D) no predicate

Q7. Short Answer

Why can an unclustered index be worse than a table scan for a low-selectivity predicate?

Q8. Calculation

A table has 90,000 tuples and 45 tuples/page. How many data pages?

Q9. MCQ

Which statement about dense and sparse indexes is true?

- (A) Dense has an entry for every search-key value/record.
- (B) Sparse can be used on any unordered heap equally well.
- (C) Sparse always has more entries than dense.
- (D) Dense indexes cannot support equality lookup.

Q10. Long Question

Recommend indexes for Student(sid, country, wam) with frequent queries by sid equality and country+wam range. Explain.



W12 Query Evaluation and Optimisation

Q1. Calculation

External merge sort on $N=1600$ pages with $B=4$ buffers: how many initial runs and total passes?

Q2. Calculation

What is the approximate I/O cost for the previous sort using $2N$ per pass?

Q3. MCQ

Which join algorithm benefits directly from having more buffer pages for the outer relation chunks?

- (A) Block nested-loop join
- (B) Tuple nested-loop join only
- (C) Natural join syntax
- (D) Projection

Q4. Short Answer

State the block nested-loop join cost formula using $B(R)$, $B(S)$, and M buffers with R outer.

Q5. Table/Diagram

Compare tuple nested-loop, block nested-loop, sort-merge and hash join in one sentence each.

Q6. MCQ

Which relational algebra rewrite is generally safe?

- (A) Push selection below a join when it references only one input.
- (B) Drop all join attributes before joining.
- (C) Replace every join by Cartesian product without condition.
- (D) Push HAVING before grouping without change.

Q7. Short Answer

Why does the optimiser need cardinality/selectivity estimates?

Q8. Calculation

Relation R has 1800 pages and relation S has 450 pages. Page nested-loop join with R as outer costs what?

Q9. MCQ

For an equality join on large unsorted inputs with enough memory to partition, which algorithm is often attractive?

- (A) Hash join
- (B) Full table Cartesian product
- (C) Repeated scalar subquery without index
- (D) Manual sorting in application code

Q10. Long Question

A query joins Employee, Assignment and Project, filters $\text{Project.budget} > 2000000$, and projects Employee.name. Give two optimisation ideas.



Answer Key

W1 Introduction

1. B.
2. True.
3. Controlled concurrency/recovery, declarative querying, integrity constraints, security, data independence; any two with explanation.
4. B.
5. Applications are insulated from logical schema changes, often through views or stable interfaces, so not every schema evolution requires application rewrites.
6. Conceptual: ER/EER model; logical: relational schema with PK/FK/constraints; physical: files, pages, indexes, access paths.
7. D. A DBMS enforces declared structures and constraints, but it does not infer all business meaning automatically. Inspired by Semester 2 2022 Final Q1.
8. Normalisation is a design-time schema refinement technique. A DBMS enforces declared constraints but does not infer and redesign normal forms for the designer.
9. $\lceil 12000/120 \rceil = 100$ pages.
10. Example: Member, Book, Author; relationships Borrows(Member,Book) with a member borrowing many books and a book borrowed by many members over time; WrittenBy(Book,Author) M:N. Other sensible answers accepted.

W2 Conceptual Database Design

1. A.
2. A weak entity cannot be identified solely by its own attributes; its key is owner key plus partial key, with an identifying relationship to the owner.
3. Customer(customerId,...); Order(orderId, customerId, ...), FK customerId; OrderLine(orderId,lineNo,qty), PK(orderId,lineNo), FK orderId.
4. C.
5. True.
6. Aggregation treats a relationship as a higher-level object so another relationship can attach to the association itself, useful when a fact is about a relationship occurrence.
7. Property strong entity; Worker supertype keyed by licenceNo; Domestic and International disjoint subtypes, likely total if every worker is one of these two; attributes qualification/visaType on subtypes. Based on Semester 2 2021 Final Q2.
8. B.
9. Because the value describes a particular pair/relationship occurrence, not either endpoint independently; putting it on one entity loses multiplicity and meaning.
10. Entities Chef, Program, Recipe; Demo Chef 1:N Program with total Program; Includes Program M:N Recipe, total Program min 1. Relations Chef, Program(programId, chefId FK), Recipe, Includes(programId, recipeId) PK pair.

W3 Relational Model and Logical Design

1. B.
2. True.
3. Bed.wno references Ward.wno; Bed can have composite key (wno,bno). Based on Semester 2 2021 Final Q3.
4. B, because AVG ignores NULL.
5. UNKNOWN OR FALSE = UNKNOWN.
6. A foreign key value referencing a candidate/primary key value, not a physical pointer.
7. Missing parent: FK violation unless FK null/not used; deleting referenced parent: violation unless cascade/set null/default; inserting parent: normally succeeds.
8. B.
9. Primary key attributes cannot be NULL and identify tuples; UNIQUE constraints may permit NULL depending on DBMS semantics and are not the chosen tuple identifier.
10. Category(cid,manufacturer); RadioSatellite(cid,frequency) FK cid; GPSSatellite(cid,imageResolution) FK cid; Technician(tid,name); Trained(cid,tid) PK(cid,tid), both FKs.



W4 Relational Algebra and SQL

1. C.
2. $\pi_{sid}(Supplier) - \pi_{sid}(Shipment)$.
3. (A=1,B=2,C=9) only.
4. C.
5. As written, the right expression has a different output schema because it keeps S attributes after the join and has no final $\pi_{A,C}$. A corrected pushed-down form would still need the final projection and must retain the join attributes. Modified from Semester 2 2024 Final Q5.
6. Join Booking to Room, then filter cost $\geq$ ALL(SELECT cost FROM Room), projecting distinct cid. Include ties.
7. Use NOT EXISTS: SELECT c.cityId FROM Cities c WHERE NOT EXISTS (SELECT 1 FROM FloodAffectedCities f WHERE f.cityId=c.cityId).
8. B.
9. Projection=SELECT list; selection=WHERE; join=JOIN/ON; difference=EXCEPT or NOT EXISTS.
10. Both are preserved; missing attributes from the other side are NULL. Student rows with no enrolment show unit/semester NULL; orphan enrolment rows show name/country NULL.

W5 Advanced SQL

1. B.
2. SELECT productId FROM Supplies GROUP BY productId HAVING COUNT(DISTINCT supplierId) $\geq$ ALL (SELECT COUNT(DISTINCT supplierId) FROM Supplies GROUP BY productId).
3. A.
4. WHERE filters rows before grouping; HAVING filters groups after aggregation, e.g. HAVING COUNT(*) $>$ 3.
5. Join Author a to Book b ON a.authorId=b.authorId; filter a.country=b.storeCountry; project DISTINCT a.name.
6. False.
7. IN tests membership in a returned set; EXISTS tests whether correlated rows exist; JOIN combines rows and can duplicate outer rows.
8. A.
9. A subquery that references attributes from the outer query, evaluated logically for each outer row.
10. COUNT(*)=3, COUNT(value)=2, SUM=20, AVG=10.

W6 Database Integrity

1. A.
2. No and no, unless the schema explicitly declares NOT NULL or UNIQUE/PK. Modified from Semester 2 2024 Final Q4.
3. A.
4. Every flight has at least one booked reservation. Modified from Semester 2 2024 Final Q6.
5. Reject/NO ACTION, CASCADE delete children, SET NULL/SET DEFAULT on child FK.
6. B.
7. True.
8. Declarative constraints are simpler, optimised by the DBMS, less error-prone, and automatically integrated with constraint checking.
9. projId=999 violates; NULL is allowed unless NOT NULL.
10. Use ON DELETE CASCADE on Employees(projId) referencing Projects(projId), or an AFTER DELETE trigger deleting matching employees; also consider business semantics and transaction atomicity.

W7 Introduction to Database Application Development

1. B.
2. Rollback the transaction so both debit and credit are undone; the business operation must be atomic.
3. UI/controller -> service/business logic -> DAO/repository -> DB driver/connection pool -> DBMS.
4. A.
5. An app loads one list then issues one extra query per row/item, causing many round trips; fix with joins, batching, or eager loading.
6. False.



7. Begin; check current enrolment/count under suitable isolation or lock; if capacity remains insert enrolment otherwise rollback. Avoid race causing over-enrolment.
8. A.
9. When an action must occur for each modified tuple after the base row change exists, such as audit rows or maintenance.
10. ROW trigger 5,000; STATEMENT trigger 1.

W8 Schema Refinement and Normalisation

1. C.
2. Any nontrivial FD must have one attribute determining the other, so the determinant determines all attributes and is a key; otherwise there is no nontrivial violation.
3. $AB \rightarrow A, B, C, D$, so AB is a key. $A \rightarrow D$ is nontrivial and A is not a superkey, so it violates BCNF. Based on Semester 2 2021 Final Q5.
4. B.
5. Update, insertion and deletion anomalies.
6. $A \rightarrow A$ starts A ; add B, C ; since C present add D ; $A \rightarrow A, B, C, D$. A is a key.
7. BCNF: every determinant is superkey; stronger, may lose dependency preservation. 3NF: allows RHS prime attribute; can be dependency-preserving.
8. A decomposition whose natural join reconstructs exactly the original relation instance, without spurious tuples.
9. B.
10. Create $R1(B, C)$ and $R2(A, B, D)$ as the standard BCNF step for $B \rightarrow C$; state lossless because intersection B determines $R1$.

W9 Transaction Management

1. C.
2. $R1(X)$ before $W2(X)$ gives $T1 \rightarrow T2$; $W1(Y)$ before $R2(Y)$ gives $T1 \rightarrow T2$ only, so acyclic and equivalent to $T1, T2$.
3. Edges include $T1 \rightarrow T2$ on x and y , $T1 \rightarrow T3$ from $W1(y)$ before $W3(y)$, $T2 \rightarrow T3$ from $R2(y)$ before $W3(y)$, and $T1 \rightarrow T4$ from $W1(x)$ before $R4(x)$. $R3(z)$ and $R1(z)$ are both reads, so no z edge. The graph is acyclic; one serial order is $T1, T2, T3, T4$. Based on Semester 2 2024 Final Q9.
4. A.
5. A transaction obtains locks during a growing phase and releases after; strict 2PL holds write/exclusive locks until commit/abort, preventing cascading aborts.
6. True.
7. Dirty read, non-repeatable read, phantom read, lost update.
8. $W1(A)$ before $R2(A)$: $T1 \rightarrow T2$. $W2(B)$ before $R1(B)$: $T2 \rightarrow T1$. Cycle, not conflict serializable.
9. A.
10. To reduce lock holding time, contention, deadlock probability, and resource usage.

W10 Quiz and Guest Lecture Integration

1. B.
2. If no nontrivial FD, BCNF holds. If $A \rightarrow B$, then A is a key; if $B \rightarrow A$, B is a key; if both, both are keys.
3. Strong entities, weak entities, multivalued attributes, M:N bridges, 1:N FKs, ISA strategy, PKs, FKs, participation/cascade notes.
4. A.
5. List all conflicting pairs on same item with at least one write in chronological order, draw directed edges, check for cycle/topological order.
6. $NULL = NULL$ gives UNKNOWN; UNKNOWN OR TRUE is TRUE.
7. Weak entities depend on their owner for identity; deleting owner usually cascades or must be rejected to avoid orphan weak rows, depending on declared FK action/business rule.
8. A.
9. If multiple rows share the maximum, a correct relational query should return all satisfying rows; use $\geq$ ALL, max join, or grouped max patterns.
10. Clarify schema and constraints, mark PK/FK/cardinalities, solve algebra with attribute checks, handle integrity actions, then transaction graph; verify NULLs, duplicates and edge cases.



W11 Storage and Indexing

- Usable=3900; effective= $0.9 \times 3900 = 3510$; floor($3510/32$)=109 records.
- $\lceil 400000/109 \rceil = 3670$ pages. Modified from Semester 2 2024 Final Q10.
- A.
- Clustered index order matches or groups data pages by index key; unclustered index points to data records/pages without physically ordering them, often causing random I/O.
- Heap: slow scan, fast insert; sorted: good range/order, expensive insert; hashed: good equality, poor range.
- A.
- It may cause many random data page fetches, potentially more expensive than sequentially scanning all pages.
- $90000/45 = 2000$ pages.
- A.
- Primary/unique B+ tree or hash on sid for equality; composite B+ tree on (country, wam) for country filtering and wam range/order; avoid over-indexing if updates heavy.

W12 Query Evaluation and Optimisation

- Initial runs ceil($1600/4$)=400. Fan-in=3. Merge passes $\lceil \log_3 400 \rceil = 6$, total passes incl pass0 = 7.
- $2 \times 1600 \times 7 = 22400$ page I/Os.
- A.
- $B(R) + \lceil B(R)/(M-2) \rceil B(S)$, assuming $M > 2$.
- Tuple NLJ simple but huge; block NLJ uses buffers; sort-merge good for equi-join/order; hash join good for equi-join if memory supports partitioning.
- A.
- To compare alternative plans and access paths by estimated output sizes and I/O costs.
- $1800 + 1800 \times 450 = 811,800$ page I/Os.
- A.
- Push budget selection to Project before joins; project only needed attrs plus join attrs; choose join order/access path using filtered Project and indexes on join keys if available.